

When Ransomware Detectors Meet Unseen Families

A strict Sysmon evaluation of rules, anomaly detectors, supervised learning, and ensembles

Anonymous submission for independent methodological feedback

Audience and scope: a specialized cybersecurity measurement paper for endpoint security and security-ML venues, written to remain accessible to general cybersecurity reviewers. ML is used for comparison, not presented as a new detector architecture.

Abstract

Endpoint ransomware detection is often reported on random train/test splits that mix behavior from the same malware family across folds. This manuscript compares rule-based detection, Isolation Forest, Local Outlier Factor, a supervised Random Forest, and two ensembles on public RADAR Sysmon telemetry. The evaluation holds out each ransomware family in turn and tests against the unseen family plus a fixed random 20% goodwill pool. On seven held-out families, Random Forest achieved macro F1 0.400 (95% CI +/-0.13) and AUC 0.809; Isolation Forest and LOF achieved F1 0.324 and 0.195. However, performance varied sharply by family, from F1 0.514 to 0.200. A 1% FPR cutoff was selected solely on inner validation: macro-average achieved test FPR was 0.005 (range 0.000-0.011) and recall was 0.166. The result supports a narrow conclusion: conventional supervised detection outperforms the tested unsupervised baselines on these RADAR windows, but performance is sensitive to unseen-family shift. It does not establish host/session-disjoint or deployment-level generalization.

Keywords

ransomware detection; Sysmon; malware-family generalization; anomaly detection; evaluation methodology; false-positive-rate calibration

1. Introduction

Ransomware detectors are commonly evaluated using endpoint or event telemetry from a finite set of observed attack families. A high score under random cross-validation can be useful for measuring in-distribution classification, but it does not answer the more operational question: does the detector retain utility when the ransomware family changes? This distinction matters because malware families differ in their staging, registry behavior, process creation, and file operations.

This study asks two limited questions. First, how do simple rules, unsupervised anomaly detectors, a supervised classifier, and small ensembles compare under a common feature representation? Second, how much does performance vary when the attack family is absent from training? The aim is not to propose a new learning algorithm. It is to make a conventional comparison auditable and to state precisely what its dataset and split can support.

1.1 Reader guide: endpoint telemetry and metrics

Sysmon is a Windows endpoint logging service that records events such as process creation, file activity, registry changes, and network connections. This study groups those events into five-minute windows and labels a window

as ransomware or benign. Precision is the fraction of alerts that are ransomware, recall is the fraction of ransomware windows detected, and F1 is their harmonic mean on a 0-to-1 scale. AUC measures how well model scores rank ransomware above benign windows across thresholds. FPR is the fraction of benign windows incorrectly alerted, and recall@1%FPR reports the attacks found after a threshold is selected for a 1% validation false-positive target. A 95% confidence interval summarizes variation across the seven held-out families; none of these values is a deployment guarantee.

2. Data and Windowed Telemetry

The study uses RADAR, a public dataset of Sysmon logs from seven ransomware families (Akira, BlackBasta, CyberVolk, LockBit, Lynx, Medusa, and Meow) and goodwill activity. Raw events are aggregated into five-minute windows. The reproducible subset contains 4,300 windows: 3,740 benign and 560 ransomware windows. Each window contains ten Sysmon-recoverable event-count features: process creation, file creation/deletion, rename-like activity, network connections, registry events, module loads, unsigned process events, suspicious extension events, shadow-copy events, and total event count.

The representation intentionally uses what the source Sysmon logs can recover. It does not claim CPU, memory, raw file entropy, or complete enterprise context. Goodware is represented by a single run; therefore, the benign test pool is a random 20% split, not a separate session-disjoint goodwill population.

2.1 Experimental system and evaluation architecture

The author did not execute ransomware or construct the RADAR laboratory environment. The experiment begins with the released Sysmon logs. Figure 1 separates the dataset provider's controlled endpoint executions from the author-controlled event aggregation, strict family split, inner validation, detector fitting, and final scoring. Family labels construct the outer split but are not predictive features. Each detector is tested once on the unseen ransomware family plus the fixed benign pool after all tuning decisions have been made without those test labels.

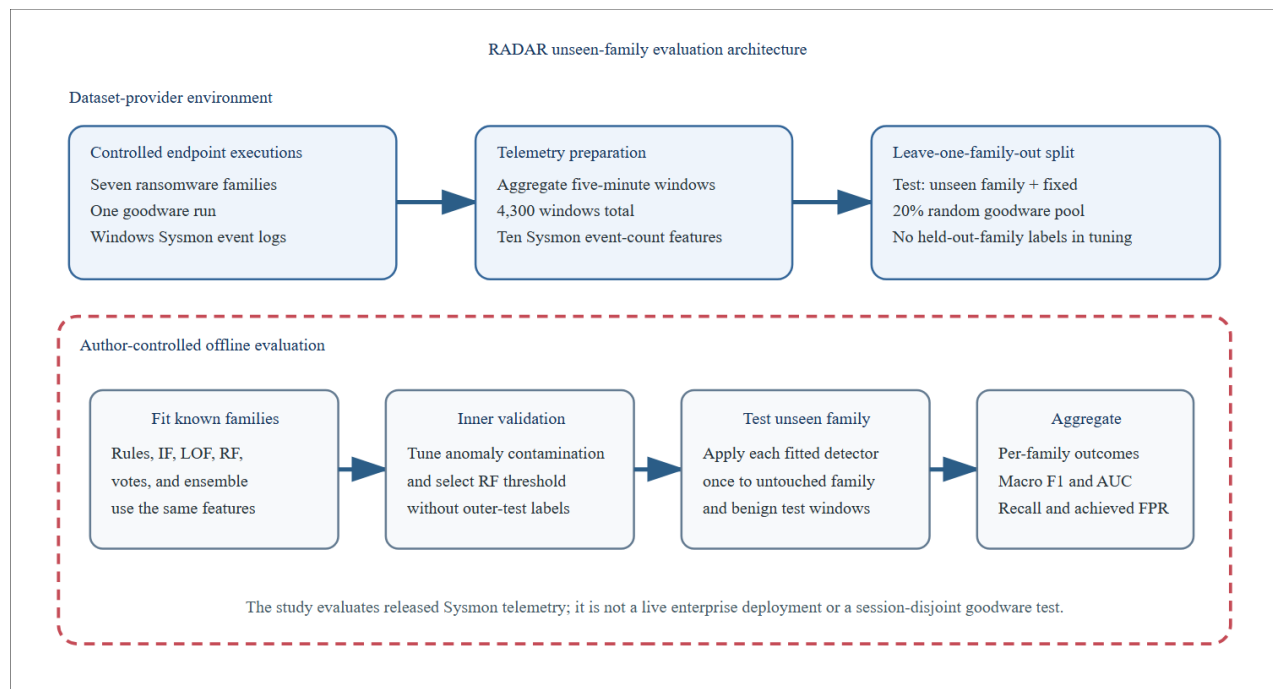


Figure 1. Dataset-provider executions and the author-controlled offline evaluation boundary.

3. Detectors and Comparators

The rule baseline sums positive indicators for rename-like activity, suspicious extensions, shadow-copy events, unsigned process activity, deletion volume, and registry volume; three indicators trigger an alert. Isolation Forest and Local Outlier Factor are trained on standardized numerical features. The supervised baseline is a class-balanced Random Forest. An unweighted vote combines Isolation Forest, LOF, and Random Forest; a weighted ensemble combines four core methods using training-fold precision weights. The progression detector is retained as a deliberately limited comparator: it requires a recon-to-tamper-to-encrypt sequence on the same host/time stream.

The goal of including these conventional methods is diagnostic. They expose different operating points: high-recall anomaly detectors, a supervised classifier whose decision boundary is learned from known families, and conservative combinations that may trade recall for precision.

4. Evaluation Protocol

The primary evaluation is leave-one-family-out. For each ransomware family, all windows from that family are withheld for testing. The test set also includes the same deterministic 20% random sample of goodware windows; the remaining goodware and all other ransomware families form the training fold. Thus, the protocol is a cross-family attack test with a held-out random benign pool. It is not a host/session-disjoint deployment test.

Isolation Forest and LOF contamination are selected only within the training fold using inner validation. Random Forest and ensemble classification thresholds remain fixed at 0.5. For the operational recall@1%FPR figure, an additional Random Forest threshold is selected on an inner validation split: among thresholds with validation FPR no greater than 1%, the one with maximal validation recall is used once on the untouched test family. The achieved test FPR is reported separately because it can vary under family shift.

- No held-out ransomware-family labels are used for contamination tuning or threshold selection.
- Metrics are macro-averaged across the seven held-out families with t-based 95% confidence intervals.
- Random cross-validation results are treated as development-era in-distribution reference results, not as the publication headline.

5. Results

Under the strict family protocol, the supervised Random Forest was the strongest individual learned detector. The unweighted vote had slightly higher macro F1, but AUC is undefined for its binary vote. The weighted ensemble was highly precise but recalled few attacks. The rules were intentionally conservative, and the progression detector did not flag RADAR trajectories because the available per-host groups were too short to establish the required three-stage sequence.

Comparator	F1 (95% CI)	Precision	Recall	AUC	Recall @ 1% FPR
Random Forest	0.400 (+/-0.13)	0.347	0.531	0.809	0.166
Unweighted vote	0.407 (+/-0.13)	0.388	0.462	-	-
Isolation Forest	0.324 (+/-0.11)	0.256	0.490	-	-
Weighted ensemble	0.208 (+/-0.06)	0.967	0.118	-	-
LOF	0.195 (+/-0.10)	0.152	0.296	-	-
Rule baseline	0.057 (+/-0.08)	0.714	0.030	-	-
Progression	0.000	-	-	-	-

Table 1. Macro-average strict leave-one-family-out results on 4,300 RADAR Sysmon windows.

How to read this table. The Random Forest row means that roughly 35% of its alerts were ransomware windows (precision 0.347) and it detected roughly 53% of ransomware windows (recall 0.531), producing F1 0.400. AUC 0.809 indicates useful but imperfect ranking. When its threshold was selected for a 1% validation FPR target, recall fell to 0.166, showing the cost of limiting false alerts.

The difference across unseen families is material. The well-sampled Akira and BlackBasta families reached F1 about 0.51, while CyberVolk and Meow reached 0.20. Small families also produce wide uncertainty. This evidence does not show that one detector will work reliably for a novel ransomware family in a live enterprise; it shows that the apparent advantage observed under random cross-validation weakens under a stricter family split.

Held-out family	Attack windows	RF F1	RF AUC	Recall @ 1%	Achieved test FPR
Akira	117	0.514	0.808	0.179	0.005
BlackBasta	128	0.511	0.829	0.141	0.000
LockBit	126	0.480	0.796	0.198	0.005
Medusa	80	0.464	0.806	0.225	0.005
Lynx	56	0.429	0.861	0.268	0.011
CyberVolk	37	0.200	0.741	0.027	0.001
Meow	16	0.200	0.820	0.125	0.009

Table 2. Per-family Random Forest outcomes. The 1% constraint is imposed on inner validation; test FPR is measured, not constrained, on the unseen family.

How to read this table. The Lynx row contains 56 held-out ransomware windows. F1 0.429 and AUC 0.861 show moderate fixed-threshold detection but stronger ranking. At the validation-selected 1% FPR operating point, recall was 0.268; the unseen-family test FPR reached 0.011, illustrating that validation calibration cannot guarantee the same false-positive rate after a family shift.

6. Interpretation

The strict results support three observations. First, labelled supervised learning remains useful within the RADAR feature representation, outperforming the individual unsupervised baselines in macro F1 and yielding an AUC of 0.809. Second, the weak rule baseline and the precision-heavy weighted ensemble illustrate why a high precision number should not be interpreted alone: both leave substantial attack activity unflagged. Third, the family-to-family variation is the main finding. Random-CV F1 of 0.515 is in-distribution optimistic compared with strict macro F1 of 0.400, and the family-specific scores make the average less reassuring than a single headline would suggest.

The calibration result should likewise be read narrowly. The threshold was selected without test labels and had macro test FPR 0.005, but one held-out family reached 0.011. This is expected under distribution shift and demonstrates why per-family achieved FPR should accompany a recall-at-budget statement.

7. Limitations

- RADAR logs are controlled laboratory executions of real ransomware and malware, not longitudinal live-enterprise telemetry.
- Malware is a single run, so the benign component is randomly held out rather than session-disjoint.
- Several ransomware families contain few windows; their point estimates have substantial uncertainty.
- The progression comparator is an omission on this source because the available traces do not support a full staged trajectory; it is not evidence that temporal detection is ineffective.
- Precision weights for the weighted ensemble are fitted on training predictions and should be treated as exploratory until evaluated with out-of-fold calibration.
- RADAR's published drift and class-imbalance scenarios are not evaluated here; this manuscript tests family holdout on the reconstructed window representation.

8. Reproducibility and Next Steps

The repository includes a raw-log adapter, a manifest of the RADAR archive, fixed seeds, strict-evaluation scripts, output JSON, dependency lockfile, and automated regression tests. The next empirical step is to evaluate on a longer-horizon endpoint corpus with genuine host/session grouping and to add RADAR's drift and imbalance settings. A second methodological improvement would calibrate ensemble weights through an inner validation or out-of-fold procedure rather than training-set precision.

8.1 Repository, data, and reproduction

Repository URL: withheld from this manuscript for double-blind review and available through an anonymous artifact when venue policy permits. Entry points: `radar_strict_eval.py` (strict leave-one-family-out evaluation), `sysmon_adapter.py` (raw-log to windowed telemetry), `real_data_eval.py` (development random-CV), plus `scripts/reproduce_radar.py` and `scripts/download_real_data.py`. Frozen preprocessing (window size, feature list, seeds, label rule) is recorded in `radar_manifest.json` and `requirements-lock.txt`.

Data attribution and license. RADAR (J. Ispahany et al., Journal of Cybersecurity, doi:10.1186/s42400-025-00435-9), Zenodo doi:10.5281/zenodo.14564541, CC BY 4.0. It is a public research dataset; the code and this manuscript are released under the repository's Non-Commercial Personal-Use License.

Reproduction commands. `python -m pip install -r requirements-lock.txt`; `python scripts/reproduce_radar.py`; `python radar_strict_eval.py --input data/radar_real_windows_with_family.csv --label-column label --split family --family-column family --metrics-output results/radar_strict_family_eval.json`; `python real_data_eval.py --input data/radar_real_windows.csv --label-column label --contamination 0.13`; `python -m pytest -q`

8.2 AI-use disclosure

AI coding assistance was used during implementation and drafting. The author directed the research question, the benchmark evaluation protocol, the strict split and recall@FPR calibration design, the interpretation of the negative family-shift result, and reviewed and verified the final code and manuscript claims. AI assistance did not set the research direction or the claims.

9. Conclusion

This work does not introduce a new ransomware detector. Its contribution is an auditable comparison showing that a familiar supervised model outperforms the tested unsupervised methods on RADAR Sysmon windows while weakening under unseen ransomware-family shift. The strict protocol and explicit limits are more important than the absolute score: the study supports cross-family evidence on a public corpus, not a deployment-level generalization claim.

References

- [1] J. Ispahany, O. B. Deho, M. R. Islam, M. A. Khan, and M. Z. Islam. RADAR: A realistic dataset for advancing ransomware detection. Journal of Cybersecurity 9(1), 2026. doi:10.1186/s42400-025-00435-9.
- [2] J. Ispahany. RADAR: Datasets, scripts and code for advancing Ransomware Detection. Zenodo, doi:10.5281/zenodo.14564541.
- [3] F. T. Liu, K. M. Ting, and Z.-H. Zhou. Isolation Forest. 2008 IEEE International Conference on Data Mining, 2008.
- [4] M. M. Breunig et al. LOF: Identifying Density-Based Local Outliers. SIGMOD Record, 2000.
- [5] L. Breiman. Random Forests. Machine Learning, 2001.
- [6] M. A. F. da Costa et al. A systematic review of ransomware detection approaches. Computers & Security, 2023.